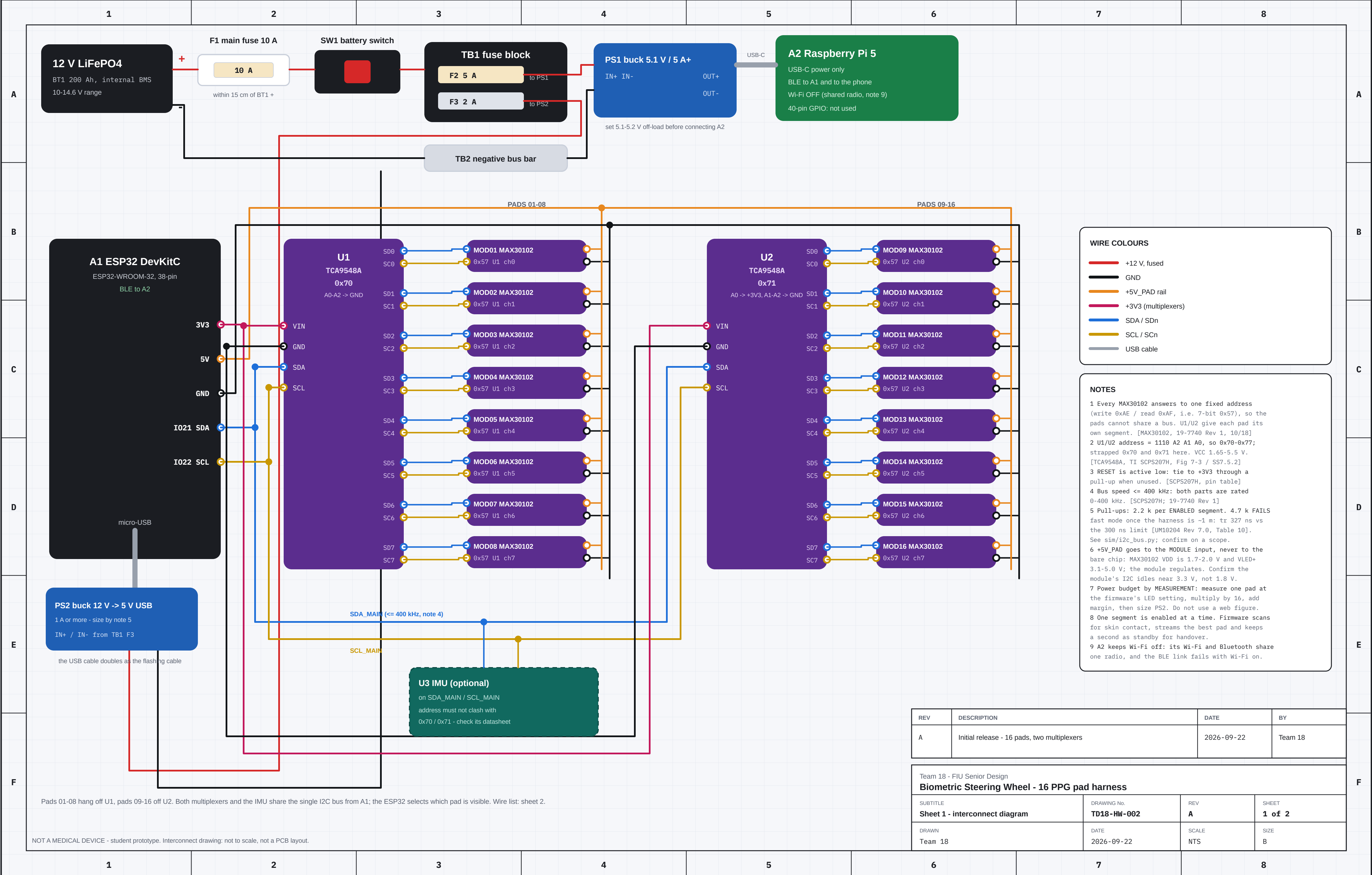




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A	PARTS LIST <table><tr><th>REF</th><th>QTY</th><th>DESCRIPTION</th><th>KEY RATINGS / NOTES</th></tr><tr><td>BT1</td><td>1</td><td>12 V LiFePO4 battery, 200 Ah, internal BMS</td><td>10-14.6 V working range</td></tr><tr><td>F1</td><td>1</td><td>Fuse + holder, 10 A</td><td>within 15 cm of BT1 +</td></tr><tr><td>SW1</td><td>1</td><td>Battery master switch</td><td>disconnects the whole system</td></tr><tr><td>TB1</td><td>1</td><td>Fused distribution block, 2 ways</td><td>F2 5 A -> PS1, F3 2 A -> PS2</td></tr><tr><td>TB2</td><td>1</td><td>Negative bus bar</td><td>single return point</td></tr><tr><td>PS1</td><td>1</td><td>Buck converter 12 V -> 5.1 V, 5 A or more</td><td>set off-load before connecting A2</td></tr><tr><td>PS2</td><td>1</td><td>Buck converter 12 V -> 5 V USB</td><td>1 A or more; size by note 5, sheet 1</td></tr><tr><td>A1</td><td>1</td><td>ESP32-WROOM-32 DevKitC, 38-pin</td><td>I2C on IO21/IO22; BLE to A2</td></tr><tr><td>A2</td><td>1</td><td>Raspberry Pi 5</td><td>USB-C power only; Wi-Fi off</td></tr></table>								REF	QTY	DESCRIPTION	KEY RATINGS / NOTES	BT1	1	12 V LiFePO4 battery, 200 Ah, internal BMS	10-14.6 V working range	F1	1	Fuse + holder, 10 A	within 15 cm of BT1 +	SW1	1	Battery master switch	disconnects the whole system	TB1	1	Fused distribution block, 2 ways	F2 5 A -> PS1, F3 2 A -> PS2	TB2	1	Negative bus bar	single return point	PS1	1	Buck converter 12 V -> 5.1 V, 5 A or more	set off-load before connecting A2	PS2	1	Buck converter 12 V -> 5 V USB	1 A or more; size by note 5, sheet 1	A1	1	ESP32-WROOM-32 DevKitC, 38-pin	I2C on IO21/IO22; BLE to A2	A2	1	Raspberry Pi 5	USB-C power only; Wi-Fi off	A
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